

NeuroGraph-X

CNN-GNN Framework for pMCI vs sMCI Prediction

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Problem Statement

- Alzheimer's progression prediction is difficult
- Key task: pMCI vs sMCI classification
- MRI = high dimensional + complex structure
- CNN $\rightarrow$ ignores inter-region relations

NeuroGraph-X

- CNN → extract brain features
- Graph → model brain connectivity
- GNN → learn disease patterns
- SVM with RBF kernel to predict probabilities of sMCI and pMCI

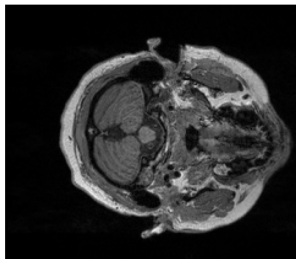
Dataset

- ADNI Dataset
- Total: 272 subjects

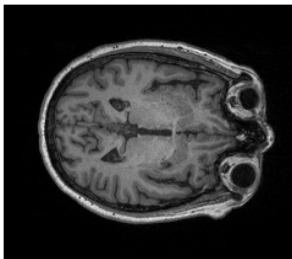
Class Distribution

- sMCI: 222
- pMCI: 50

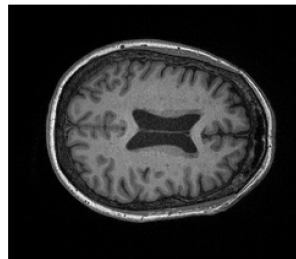
Slice 70



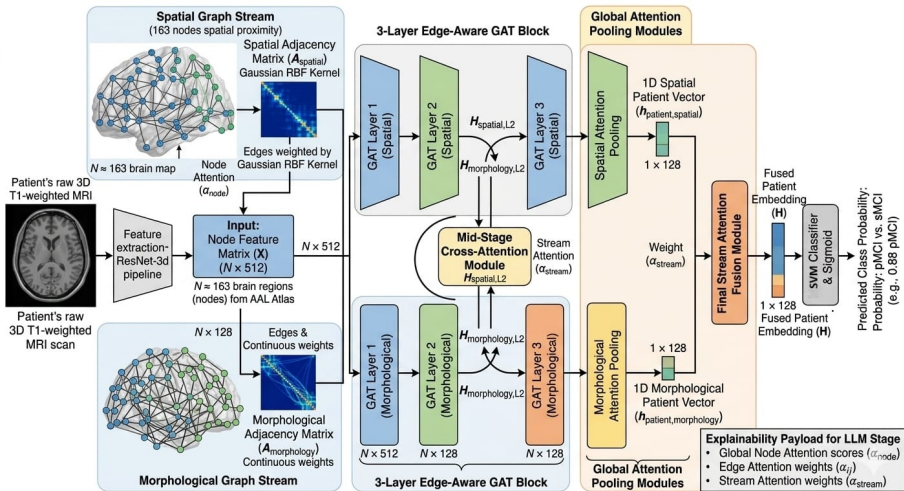
Slice 128



Slice 150



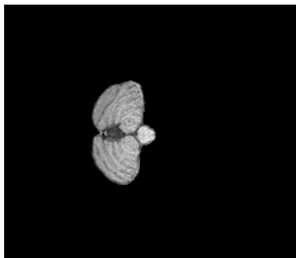
Pipeline Overview



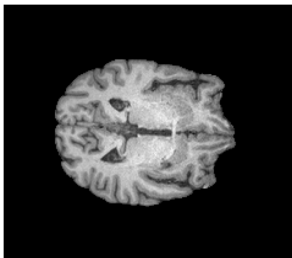
Data Pipeline

- Skull Stripping (HD-BET)
- Bias Correction (N4)
- Registration (MNI)
- Normalization

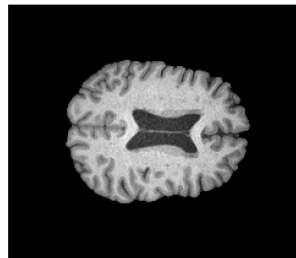
Slice 70



Slice 128



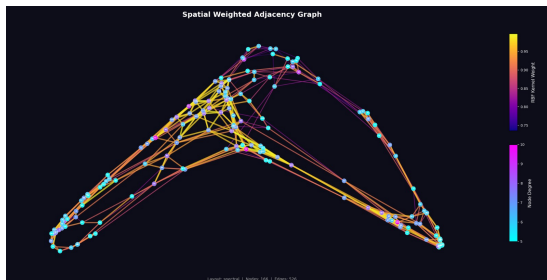
Slice 150



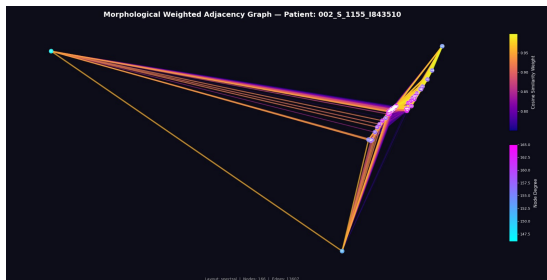
Brain Graph Construction

Spatial Graph

- KNN connections
- Distance-based edges
- RBF weighting



Brain Graph Construction



Morphological Graph

- Cosine similarity
- Adaptive threshold
- Feature-based edges

Feature Extraction

- 3D ResNet → regional features
- Global CNN → whole brain context

Output

- Node features: 166×512
- Global feature: 256-dim

GNN Architecture

- Dual-stream GATv2
- Edge-aware attention
- Multi-head aggregation

Innovation

- Uses edge weights in attention
- Parallel spatial + morphological learning

Cross-Attention Fusion

- Spatial Morphological interaction
- Information exchange between streams

Benefit

- Captures complex brain relationships
- Improves representation quality

Final Classification

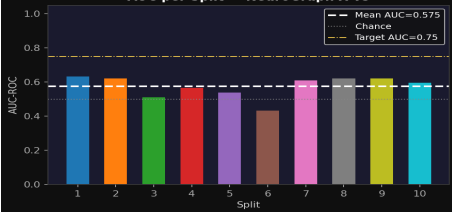
- Attention pooling $\rightarrow$ graph embedding
- Fusion with global CNN features

Classifier

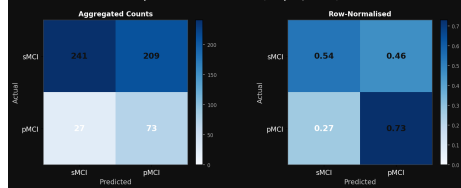
- SVM (RBF kernel)
- Better generalization (small dataset)

Results

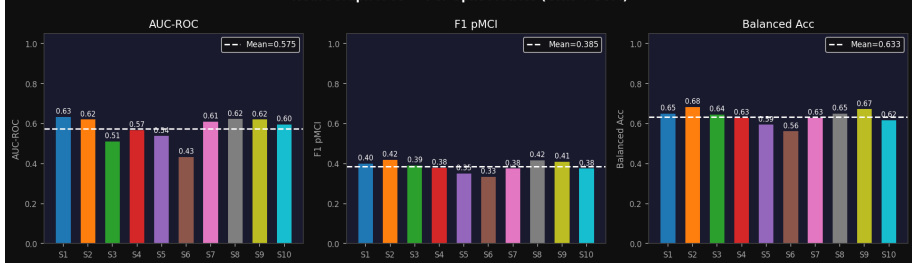
AUC per Split — NeuroGraph-X v5



NeuroGraph-X v5 — Confusion Matrix (10 splits, N=550)

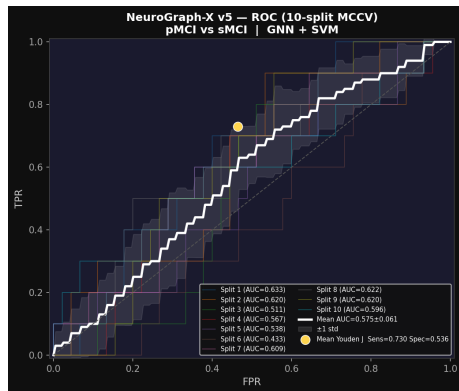


NeuroGraph-X v5 — Per-Split Metrics (GNN + SVM)



Results

- AUC: 0.57
- Bal Acc: 0.63
- Recall (pMCI): 0.73
- Specificity: 0.53



Comparison with SOTA

Model	AUC	Acc	Recall
VGG-TSwinformer	0.81	0.77	0.79
Hippocampal ViT	0.74	0.69	0.65
Ours	0.59	0.63	0.73

Key Insight

- High recall → strong early detection
- Trade-off: lower specificity

Conclusion

- Combines CNN + GNN
- Captures brain connectivity

Future Work

- Add multimodal data (PET)
- Improve specificity
- Clinical deployment

References



Jun Cao *Et al.* Dementia classification using a graph neural network on imaging of effective brain connectivity. *Computers in Biology and Medicine*, 168:107701, 2024.



Yinghua Fu *Et al.* and Alzheimer's Disease Neuroimaging Initiative. Alzheimer's disease classification using mutual information generated graph convolutional network for functional mri. *Journal of Alzheimer's Disease*, 106(3):1021-1035, 2025.



Zhentao Hu *Et al.*. VGG-TSwinformer: Transformer-based deep learning model for early Alzheimer's disease prediction. *Computer Methods and Programs in Biomedicine*, 229:107291, 2022.



A. Balasundaram, Sruthi Srinivasan, A. Prasad, Jahan Malik, and Ayush Kumar. A Hippocampal-Based Hybrid Transformer Network for Early Prediction of Alzheimer's Disease. *Arabian Journal for Science and Engineering*, 48:10521–10534, 2022. Available: <https://link.springer.com/article/10.1007/s13369-022-07538-2>

Thank You